

Do AI Dream of Real Images: An Evaluation of the Capability of AI to Recognise AI-generated images

Author: Dillon Adli Gunawan, Supervisor: Prof. Irwin King

The Chinese University of Hong Kong

Abstract

The rapid advancement of generative artificial intelligence (GenAI) has led to the creation of highly realistic synthetic images, raising concerns about misinformation, fraud, and security threats. This has only worsened as GenAI architectures developed, such as Generative Adversarial Networks (GANs) and Latent Stable Diffusion models, have made it more difficult for the human eye to distinguish between real or fake images. However image classification models, particularly Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), have shown promise in distinguishing between real and AI-generated images. The objective of this study is to evaluate the performance of these deep learning models against each other to observe which will perform the best. Our method to do this is by fine-tuning commonly used image classification models to detect AI-generated image. This is done by having multiple training datasets, with one training dataset for each GenAI architecture, and multiple model types, one for each type of image classification architecture. Then, one dataset and one model type is paired to create one trained model. We additionally make use of a hybrid dataset, consisting of generated images from multiple generator architectures, to observe whether there are artefacts universal to all generators that can be detected. Finally, all the trained models, from all possible dataset-model pairings, are compared on their performance on the same test dataset based on various metrics, such as accuracy, false positive rate, false negative rate and etc. Our experiments indicate that CNNs are the best at detecting generated images, with the hybrid dataset allowing for more generalized performance across various generator types.

1 Introduction

With the recent emergence and development of AI, one of the major uses of this technology is the creation of AI-generated content, specifically images. Images can now be quickly generated in a matter of seconds, and the variety of content produced can be diverse. Furthermore, the accessibility of this capability has only grown, with cheaper machines and online websites that can create such content. This

has essentially enabled anyone on the internet to create images of anything they want at any time for little to no cost on their own.

This, like any newly introduced technology, has led to many use cases, some good, such as for historical reconstruction of ancestors, heritage sites or ancient cities, and the creation of X-ray and MRI images for use as a dataset for diagnosis algorithms, but more relevant to this paper would be unethical uses. For example, one of the most prevalent uses is deepfakes. Deepfakes are images or videos, which another person's face or body can be digitally altered to look like another person. There are frauds and scams, in which a famous person's likeness, such as Elon Musk, has been used to prompt fraudulent investment opportunities, causing an estimated 12 billion U.S. dollars in losses in just a year (New et al. 2024), and disinformation being spread, like during the 2024 U.S. presidential election, where famous figures and celebrities are deepfaked to falsely endorse a presidential candidate (Bond 2024). Of course, impersonating another person digitally has always been an issue due to the use of photo/video-editing software, but AI has made it more rampant, and easy access to image generation has created new avenues of unethical behavior. An example of unethical use was when Midjourney, one of the most popular generative AI software, made AI art that was indistinguishable from human art, resulting in it winning the Colorado State Fair Fine Arts competition (Metz 2022).



Figure 1: The generated image, named "Théâtre D'opéra Spatial", which the digital arts/digitally-manipulated photography category of the Colorado State Fair Fine Arts Competition (Metz 2022)

This used to be not such a big issue, as in the past, generative AI was still primitive compared to today’s technology, there were obvious visual defects and errors in the generated images, or they had a certain recognizable style that was easy to differentiate from real images. But with the development of GAN (Generative Adversarial Network), Stable Diffusion, and Transformer architecture for generative AI, today’s artificial content has become harder and harder for the human eye to identify and differentiate. Hence, these unethical uses and potential harm have led to the growing importance of detection methods for the classification of AI images. To keep up with the advancing technology, many studies and works were done to figure out ways to classify AI images. One of the most promising developments involved the use of deep learning models used for image classification, and fine-tuning them using real and AI images for them to be able to distinguish features or artefacts unique to AI images, with the two most commonly used architectures being CNNs (Convolutional Neural Networks) and ViTs (Vision Transformers). However, there is a lack of literature in directly comparing these two architectures and their respective models when it comes to their accuracy and efficacy in detecting AI-generated content.



(a) Generated image in 2022 (b) Generated Image in 2025

Figure 2: Two images generated in 2022 and 2025 respectively, with the same prompt "An elementary school class photo of dogs". (Magee 2025)

Therefore, the main goal of this paper is to evaluate and compare the performance of commonly used image classification models. The main contributions to this goal are:

- The collection and creation of datasets, which will then be used to fine-tune commonly used image classification models, into binary classifiers for AI image detection
- The evaluation of the trained models against each other on the same test set on a variety of metrics, which will be collected in a table below
- The comparison of models trained on generated images of one architecture tested on generated images of another architecture (i.e. A GAN-trained CNN tested on detecting diffusion-generated images), and evaluating the possibility of a universal detector

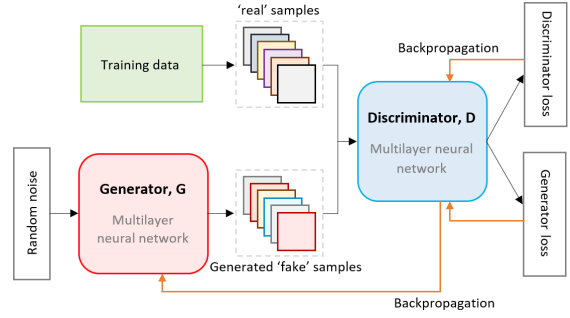


Figure 3: General Architecture of a General Adversarial Network (Little et al. 2021)

2 Related Works

2.1 Generative Adversarial Networks (GANs)

General Adversarial Networks (Goodfellow et al. 2014) is a deep learning architecture that consists of two main parts, the generator and the discriminator. The generator is a multilayer perceptron with the purpose of generating content. It does so by taking some Gaussian noise as input (which can be defined as z), and converting the noise into an image. Conceptually, the noise represents the features of the resulting image, such as object type, size or color. This function of the generator can be represented as $G(z)$. A generator at first will produce random images, and will continue to do so without any external change, but that is the purpose of the discriminator. The discriminator is another multilayer perceptron with the purpose of receiving the generator image from the generator or a real image dataset and deducing whether it is real or not. (This can be represented by a probability function $D(x)$, where x can be an image from either the generator or a real dataset, outputting a continuous value in the range of 0 and 1, representing the probability that the input image is real). The generator is then trained by trying to minimize the probability that the discriminator thinks that the image is fake, thereby resulting in more and more realistic images as the training process is done. The discriminator is also trained using both a real dataset and generated images. This architecture is essentially a two-player game between the generator and discriminator, where the generator and discriminator are both improving in order to generate realistic-looking images, and detecting them with greater accuracy, respectively.

Hence the general architecture of a GAN can be defined as the following equations:

$$\min_G V(G) = \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))].$$

An equation representing the Generator minimizing the chance the generated image is recognized as fake

$$\max_D V(D) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))]$$

An equation representing the Discriminator maximizing the chance real images are recognized as real and generated images are recognised as fake

$$\min_G \max_D V(D, G) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))]$$

The overall min-max 'game' that represents the architecture of a GAN

2.2 Stable Diffusion

The general process in which a diffusion model generates an image can be described as the model receiving a noisy image and gradually denoising the image to eventually receive the desired image (Ho, Jain, and Abbeel 2020). The model learns how to denoise a noisy image, also known as backwards diffusion, by first, being trained by doing forward diffusion. Forward diffusion is when noise is gradually added to an image of multiple timesteps, which will allow the model to learn training parameters to restore a noisy image to the state of its previous timestep, and eventually get a realistic image, therefore achieving backwards diffusion. Originally, diffusion was done in the pixel space (Ho, Jain, and Abbeel 2020), which caused long execution times, but later, a Latent Diffusion Model was developed (Romach et al. 2022), allowing for the diffusion process to be done in the latent space. This also allowed for text prompts to be more easily used for image-to-text generation, as embeddings in the latent space can be paired with images, allowing for conditional diffusion to occur.

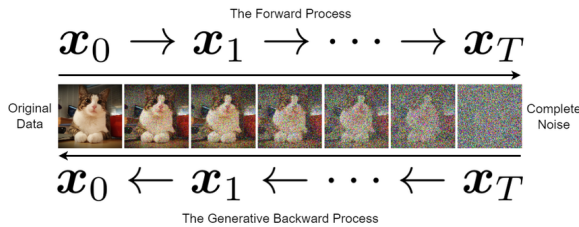


Figure 4: The process of adding noise and denoising for forward and backward diffusion. (Ghojogh and Ghodsi 2024)

2.3 Convolutional Neural Networks (CNNs)

One issue for using a typical multilayer perception for image classification is the size of the input into the model. For example, assuming a full-colour 1000 x 1000 image, since each pixel consists of three colours (red, green, blue), will need to pass 3 000 000 data points, and therefore need 3 000 000 nodes in the input layer, to fully comprehend the image, and much more in the hidden layers, hence a very high re-

source requirement is needed for the model. Convolutional Neural Networks (Phung and Rhee 2019) mitigate this issue by making use of a convolutional layer. The convolutional layer consists of filters, known as kernels, that is applied to the original image. From a high-level perspective, these kernels are enabling the extraction of higher-level features, such as edges, texture, lines, curves, and etc.

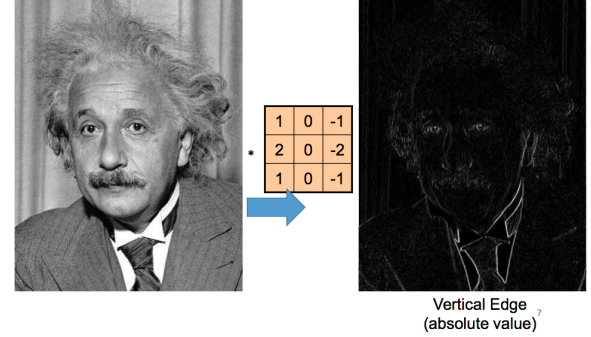


Figure 5: An example of a kernel being multiplied to an image to identify vertical lines. (Zhang 2025)

Then the resulting data from applying filters are passed to the pooling layers. These pooling layers take a section of the resulting image, and do max pooling on them (taking the highest value in a section and passing that value), and do so to extract the most significant information from the section. There can be multiple convolution and pooling layers to further extract higher-level features (i.e. lines and edges from pixels, shapes from edges and lines), and at the end of the convolution and pooling layers, the resulting data will be passed to the multilayer perceptron to classify the image. As the original image is passed through convolutional and pooling layers, the amount of data passed to the multilayer perceptron will therefore be greatly reduced.

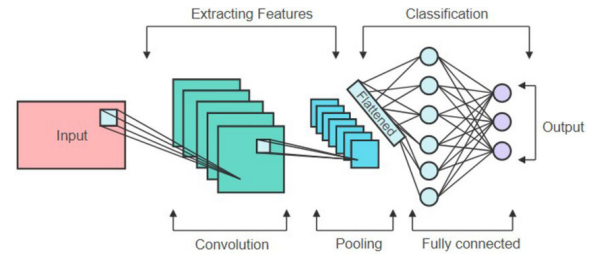


Figure 6: General architecture of a Convolutional Neural Network. (Nagy et al. 2023)

2.4 Vision Transformers (ViTs)

A Vision Transformer (He et al. 2016) is an encoder-only transformer with the purpose of image classification. It splits an image into equally sized patches, and then converts these patches into vectors, known as embeddings. These embeddings are also given their own positional encoding, to represent the position of their respective patch in the image, to allow for the attention head to understand the connection

between the matches. A special embedding, known as the CLS, is used as the classification token and is usually put as the first embedding at 0th position. Then all the embeddings are passed into the transformer encoder. In the encoder it will pass through an attention head, where using queries, keys, and values, the connection and ‘context’ of the embedding to each other will be synthesised and used to update the value of the patches. After that, the embedding will then be passed onto a multilayer perceptron where a nonlinear transformation is applied, allowing for better feature extraction. After passing through the transformer encoder, the CLS is then passed to a multilayer perceptron to classify what kind of image it is.

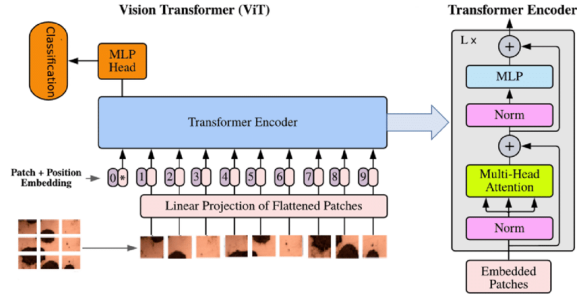


Figure 7: General architecture of a Vision Transformer. (Alfou, Kordijazi, and Zhao 2022)

2.5 Models

For the image classification models we use, we make use of pre-trained models and weights, as it will allow for already learned features and patterns, especially for real images, to be used in detecting AI-generated images.

- For CNNs, a pre-trained ResNet50 based on the study done by (He et al. 2016) will be used. The difference between ResNet50 and other CNNs is that it makes use of residual learning, where residual blocks are used to allow for gradients to flow through the network, and prevent vanishing gradients during backpropagation, which allows for models to have greater depth without degrading. The pre-trained weights are trained by using ImageNet images to classify 1000 classes, on 1.3 million training images. The model is trained on 224×224 pixels-sized images.
- For ViTs, a pre-trained Vision Transformer based on this study done by Dosovitskiy et al (2020) (Dosovitskiy et al. 2020). The pre-trained model is trained on the ImageNet-1k dataset to classify 1000 classes, using 1.3 million images. The model is trained on a resolution of 224×224 pixels with a patch size 16×16 pixels.

Furthermore, since these pre-trained models are multi-class image classifiers, they are converted into binary classifiers by replacing the final layer with a fully-connected layer with one output neuron for the CNN, and replacing the classification head with a single layer with a ReLU and one output neuron. Additionally, a lower learning rate will be used to help retain important weights and learned feature detection.

We do not freeze the non-final layers and neurons, to still allow for the better learning of the features of AI-generated images, as the pre-trained models were not trained on them and it will be essentially a new type of class of images to them.

2.6 Dataset Source

- From this CNN detection study (Wang et al. 2020), GAN-generated images and real images are provided. These images are generated based on 20 different classes. Specifically the GAN-generated images are generated from a variety of GAN models.
- From this diffusion detection study (Corvi et al. 2023), 200k Latent Diffusion Model-generated images. The images are generated based on the COCO (Common Objects in Context) and LSUN (Large-scale Scene Understanding) datasets and their respective classes found in the datasets.

2.7 Training Dataset

For training datasets, we made three datasets: a GAN, a Diffusion, and a Hybrid Dataset. All the datasets comprise of a 50-50 split of real images and generated images. The main difference between the datasets is the composition of the type of generated images. The generated images found in the GAN dataset will only contain GAN-generated images, while the generated images found in the Diffusion dataset, will only contain Diffusion-generated images. However, the generated images found in the Hybrid dataset will comprise 50% GAN-generated images, while the other 50% will contain Diffusion-generated images, but the number of generated images will remain consistent with the other datasets.

- For the GAN dataset, 5000 generated images and 5000 real images, totalling to 10000 images for the training set. The GAN-generated and real images are taken from the CNN detection study.
- For the diffusion dataset, 5000 generated images and 5000 real images, totalling to 10000 images for the training set. The diffusion images are taken from the diffusion detection study, while the real images are taken from the CNN detection study.
- For the hybrid dataset, 5000 generated images, 2500 GAN-generated and 2500 diffusion-generated, and 5000 real images, totalling to 10000 images for the training set. The Diffusion images are taken from the diffusion detection study, while GAN-generated and real images are taken from the CNN detection study.



Figure 8: Sample GAN-generated images from training dataset

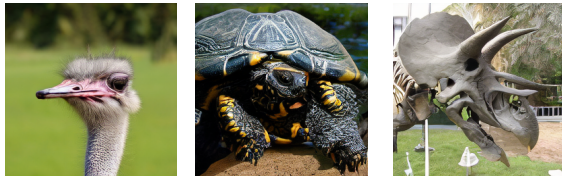


Figure 9: Sample Diffusion-generated images from training dataset



Figure 10: Sample real images from training dataset

2.8 Test Dataset

For test datasets, it will consist of two datasets: a GAN dataset and a Diffusion dataset. Like the training datasets before, the main difference of the dataset will be the origin architecture of the generated images. The reason we do not have a hybrid dataset for testing is that we can combine the results achieved from the GAN and Diffusion datasets and combine them to see their overall performance, effectively mimicking a hybrid dataset, while still being able to see the specific performance of each model based on a specific generator architecture.

- For the GAN dataset, 1000 generated images and 1000 real images, totalling to 2000 images for the training set. The GAN-generated and real images are taken from the CNN detection study.
- For the diffusion dataset, 1000 generated images and 1000 real images, totalling to 20000 images for the training set. The diffusion images are taken from the diffusion detection study, while the real images are taken from the CNN detection study.
- Hence, the overall/hybrid dataset will be compared based on 2000 generated images, 1000 GAN-generated and 1000 diffusion-generated, and 2000 real images.



Figure 11: Sample GAN-generated images from testset



Figure 12: Sample Diffusion-generated images from testset



Figure 13: Sample real images from testset

2.9 Evaluation

Evaluation will be done based on three metrics: Accuracy, False Positive, False Negative percentage.

- Accuracy (Acc): The percentage of 'correct' classification, i.e. real images being flagged as real, and AI-generated being flagged as generated.
- Real Accuracy (Real Acc): The percentage of real images being classified as real images
- AI Accuracy (AI Acc): The percentage of AI images being classified as AI images
- False Positive (FP): The percentage of real Images flagged as AI-generated
- False Negative (FN): The percentage of AI-generated images flagged as real

Aside from these metrics, a GradCAM (Gradient-weighted Class Activation Mapping) will be used to observe the areas in which the areas and patterns on an image to identify whether it is fake or real. GradCAM is a technique designed to make CNNs more interpretable by highlighting the regions of an image that influence the model's predictions. Unlike traditional methods that treat deep learning models as black boxes, Grad-CAM provides visual explanations, allowing us to further understand which parts of an input image the model considers important to differentiate real and generated images.

2.10 Experiment Setup

Using the 3 datasets, and 2 models, a pair of each dataset and each model type will be made, so there will be 6 unique pairings. In each pairing the model is trained on the dataset to create one final unique model. Therefore, the 6 unique models that are fine-tuned are:

- GAN-Trained CNN (CNN GAN)
- Diffusion-Trained CNN (CNN Diff)
- Hybrid-Trained CNN (CNN Hyb)
- GAN-Trained ViT (ViT GAN)
- Diffusion-Trained ViT (ViT Diff)

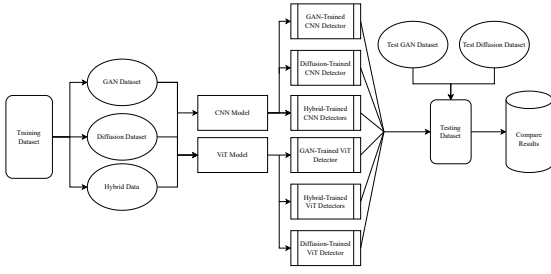


Figure 14: Workflow of the experimental setup

- Hybrid-Trained ViT (ViT Hyb)

These models are then tested based on the 2 test datasets, and evaluated based on the metric previously mentioned.

3 Results

Comparison results based on the evaluation metrics, involving the aforementioned metrics, will be displayed in the three tables below. Each table will represent the performance of the 6 models on each of the test datasets, and their overall performance, calculated by summing and combining all the other datasets. In the table each columns represent a metric, and the numbers within those columns is are percentage values.

3.1 Evaluation on GAN Images

	Accuracy	Real Acc	Ai Acc	FP	FN
CNN GAN	98.1	99.4	96.8	0.6	3.2
CNN Diff	61.1	98.8	23.4	1.2	76.6
CNN Hyb	99.8	100	99.6	0	0.4
ViT GAN	89.2	97.4	81	2.6	19
ViT Diff	49.1	97.4	0.8	2.6	99.2
ViT Hyb	90.1	89	91.2	11	8.8

The results of the GAN-trained models indicate that GAN images are very easily detected, with all models achieving at least above 90% accuracy when identifying GAN-generated images. Furthermore, for GAN-trained models the CNNs were performing with much higher accuracy than their ViT counterparts, with nearly 10% differences between them. When viewing the accuracy of the Diffusion-trained models, it is clear that their accuracy is far worse, indicating that the artefacts found in diffusion-generated images are different and unapplicable to detecting GAN-generated images. Additionally, the Models trained on the hybrid dataset show similar trends to GAN-trained models, with even the hybrid-trained CNN performing slightly better than its GAN-trained counterpart. Though the difference between the GAN and Hybrid-trained models is too minor, and with the fact that the Hybrid-trained ViT performing worse than the GAN-trained one, it is not an indication that Hybrid-trained or GAN-trained models perform better than the other, rather that they have similar performances. This is despite the hybrid training set containing diffusion images, possibly 'poi-

soning' the model, but instead still maintaining great accuracy. Overall, all the models were still able to correctly identify real images as real; however, it is false negatives that are being made, contributing to the inaccuracies of the worse-performing models.

3.2 Evaluation on Diffusion Images

	Accuracy	Real Acc	Ai Acc	FP	FN
CNN GAN	70.7	99.6	41.8	0.4	58.2
CNN Diff	95	99.2	90.8	0.8	9.2
CNN Hyb	93.8	99.6	88	0.4	12
ViT GAN	51.1	98	4.2	2	95.8
ViT Diff	87.8	97.4	78.2	2.6	21.8
ViT Hyb	83.3	89.2	77.4	10.8	22.6

The performance on the Diffusion test datasets shows similar trends to the GAN dataset, with the Diffusion or Hybrid-trained models performing similarly, while the models trained on a different generator architecture, the GAN-trained models, performed the worst. However, the overall accuracy on models trained on the same generator type is worse for Diffusion images compared to GAN images, which could indicate greater difficulty when identifying Diffusion-generated images. Other than that, ViTs also perform worse than their CNN counterparts again. Additionally, the GAN-trained CNN, though performing much worse than the Diffusion-trained CNN, still achieved 70% accuracy, much better than the 60% achieved by the Diffusion-trained CNN tested on a GAN dataset, which could indicate that artefacts found in GAN-generated images being more universal than Diffusion-generated images. Furthermore, once again the inaccuracies can be mostly attributed to False negatives being made, as Real accuracy remains high.

3.3 Evaluation on overall performance

	Accuracy	Real Acc	Ai Acc	FP	FN
CNN GAN	84.4	99.5	69.3	0.5	30.7
CNN Diff	78.05	99	57.1	1	42.9
CNN Hyb	96.8	99.8	93.8	0.2	6.2
ViT GAN	70.15	97.7	42.6	2.3	57.4
ViT Diff	68.45	97.4	39.5	2.6	60.5
ViT Hyb	86.7	89.1	84.3	10.9	15.7

In the overall comparison, it can be seen that the accuracy of models are all at least above 60%, however this is probably due to their Real accuracy being very high, with all models achieving nearly 90% accuracy or greater, and AI accuracy being where most mistakes in detection were made. It can also be seen that the Hybrid-trained models are superior to the more specialised-trained models, achieving greater accuracy when faced with all types of generated images. It can also be seen that in general, the CNN models performed with greater accuracy than the ViT models, with a much lower False Negative rate.

4 GradCAM Results

Furthermore, we also decided to utilise a GradCAM on the CNN models, to further investigate the difference in the features considered important between the different CNN Models, by observing the generated heatmaps, from correctly detected fake AI images.

4.1 GAN GradCAM Results

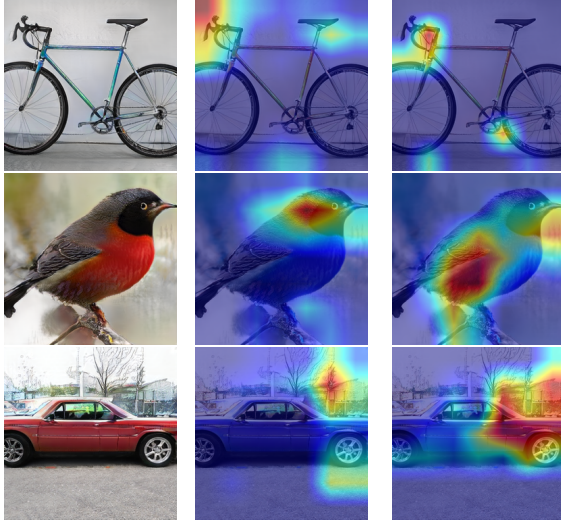


Figure 15: GAN images and their generated heatmaps. (Left column is the original images, middle column is the heatmap generated by GAN-trained CNN, and right column is the heatmap generated by Hybrid-trained CNN.)

As can be seen from the images above, the heatmaps generated from a GAN-trained CNN and a Hybrid-trained CNN are significantly different. The general location of the hotspots found in the GAN-trained can be found in the Hybrid-trained CNN too, though the specific size and location of these 'shared' hotspots can differ. The main difference between the GAN-trained and Hybrid-trained heatmaps is the fact that the Hybrid-trained heatmaps contain hotspot regions that cannot be found in the GAN-trained ones. Furthermore, these hotspots are generated and are not just small differences, but are actually quite large, even having a similar size to the 'shared' hotspots.

4.2 Diffusion GradCAM Results

Similarly to the GAN-generated images, the heatmaps for the diffusion generated images, made using the Diffusion-trained CNN and the Hybrid-trained CNN contain some differences, though it can be seen that the differences are less significant. Similarly to the GAN images, the hotspots in the Diffusion-trained heatmaps are also 'shared' with the Hybrid-trained heatmaps. However, unlike the GAN images, there are no heat regions found in the Hybrid-trained heatmaps, that cannot be found in the Diffusion-trained heatmaps; rather the Diffusion-trained heatmaps have more concentrated hotspots, while the Hybrid-trained heatmaps

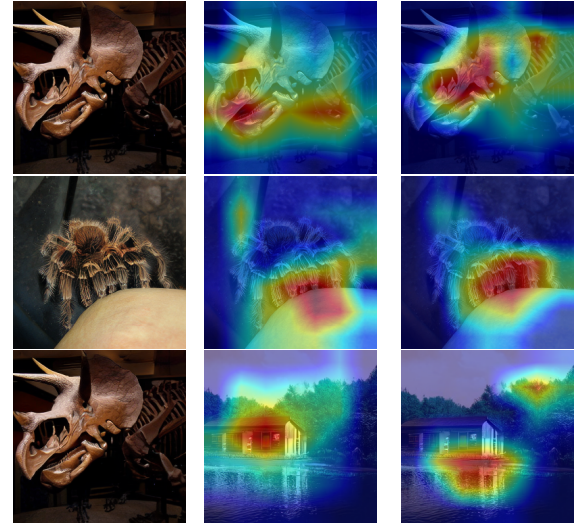


Figure 16: Diffusion images and their generated heatmaps. (Left column is the original images, middle column is the heatmap generated by Diffusion-trained CNN, and right column is the heatmap generated by Hybrid-trained CNN.)

have larger hotspots, with the less 'hot' regions in the Diffusion-trained ones being red in the Hybrid-trained ones.

Overall, it can be seen that the important features of a Hybrid-trained model are significantly different to the ones trained on a specific generator architecture. Furthermore, a hybrid-trained model will lead to a wider range of features to be detected, while still maintaining similar performance.

5 Conclusion and Future Work

Overall, it can be seen that current deep learning image classification models are able to identify and detect AI-generated images. Furthermore, it can also be seen that different generator types, have different artefacts to be detected, and that a detector trained on one type will have difficulties detecting other types of generated images. To overcome this issue, training detectors on hybrid datasets has shown to maintain similar performances on specific individual generated image types compared to more models trained specifically on that generator type, but still being able to detect other types of AI images. Furthermore, it can be seen that CNNs perform better than ViTs in AI image detection.

Detection of AI-generated images is an ever-evolving issue, with the development and improvement of both image generators and classification models. Hence, our next research study will conduct experiments on a greater number of images, with a specific focus on differing subtypes of each architecture, such as the different type of GAN generators, such as CycleGAN or BigGAN.

References

Alrfou, K.; Kordijazi, A.; and Zhao, T. 2022. Computer Vision Methods for the Microstructural Analysis of

Materials: The State-of-the-art and Future Perspectives. arXiv:2208.04149.

Bond, S. 2024. How AI deepfakes polluted elections in 2024. National Public Radio. [Online; accessed 2025-07-21].

Corvi, R.; Cozzolino, D.; Zingarini, G.; Poggi, G.; Nagano, K.; and Verdoliva, L. 2023. On the detection of synthetic images generated by diffusion models. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 1–5.

Dosovitskiy, A.; Beyer, L.; Kolesnikov, A.; Weissenborn, D.; Zhai, X.; Unterthiner, T.; and Houlsby, N. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*.

Ghojogh, B.; and Ghodsi, A. 2024. Diffusion Models: Tutorial and Survey.

Goodfellow, I. J.; Pouget-Abadie, J.; Mirza, M.; Xu, B.; Warde-Farley, D.; Ozair, S.; and Bengio, Y. 2014. Generative Adversarial Nets. In Ghahramani, Z.; Welling, M.; Cortes, C.; Lawrence, N.; and Weinberger, K. Q., eds., *Advances in Neural Information Processing Systems*, volume 27, 2672–2680.

He, K.; Zhang, X.; Ren, S.; and Sun, J. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, 770–778.

Ho, J.; Jain, A.; and Abbeel, P. 2020. Denoising Diffusion Probabilistic Models. In Larochelle, H.; Ranzato, M.; Hadsell, R.; Balcan, M. F.; and Lin, H., eds., *Advances in Neural Information Processing Systems*, volume 33, 6840–6851.

Little, C.; Elliot, M.; Allmendinger, R.; and Samani, S. 2021. Generative Adversarial Networks for Synthetic Data Generation: A Comparative Study. *arXiv preprint*.

Magee, Z. 2025. How Far We’ve Come: A Comparison Of AI Generated Images And What It Means For You. Accessed: 2025-07-28.

Metz, R. 2022. AI won an art contest, and artists are furious. CNN. [Online; accessed 2025-07-21].

Nagy, W.; Alsalamah, H.; Hassan, M.; and Moahamed, E. 2023. AUTO-HAR: An adaptive human activity recognition framework using an automated CNN architecture design. *Heliyon*, 9: e13636.

New, B.; Salazar, L.; Lozano, M.; and Fralicks, S. 2024. Deepfakes of Elon Musk are contributing to billions of dollars in fraud losses in the U.S. CBS News. [Online; accessed 2025-07-21].

Phung, V.; and Rhee, E. 2019. A High-Accuracy Model Average Ensemble of Convolutional Neural Networks for Classification of Cloud Image Patches on Small Datasets. *Applied Sciences*, 9(21): 4500.

Rombach, R.; Blattmann, A.; Lorenz, D.; Esser, P.; and Ommer, B. 2022. High-Resolution Image Synthesis With Latent Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 10684–10695.

Wang, S. Y.; Wang, O.; Zhang, R.; Owens, A.; and Efros, A. A. 2020. CNN-generated images are surprisingly easy to spot... for now. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 8695–8704.

Zhang, L.-C. 2025. Convolutional Neural Networks Lecture Notes. <https://www.cs.toronto.edu/~lc Zhang/360/lec/w04/convnet.html>. Accessed: 2025-07-28.